[INCH-POUND]
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December 13, 1999
SUPERSEDING
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October 27, 1982

COMMERCIAL ITEM DESCRIPTION

SHOE, MOLDERS

The General Services Administration has authorized the use of this commercial item description for all Federal agencies.

1. SCOPE. This commercial item description covers the requirements for materials and manufacture of a Molders style work shoe with safety toe and oil resistant soles and heels used by personnel of the Department of Defense.

2. CLASSIFICATION. The shoes shall be of one type in the following whole and half sizes and width as specified.

Sizes	Widths
4 to 14-1/2	XN - Extra narrow
4 to 14-1/2	N - Narrow
4 to 14-1/2	R - Regular
4 to 14-1/2	W - Wide
4 to 14-1/2	XW - Extra wide

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, Attn: DSCP-CR, 700 Robbins Avenue, Bldg. 6D, Philadelphia, PA 19111-5096.

AMSC N/A FSC 8430

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3. SALIENT CHARACTERISTICS

3.1 Description. The design shall be a 6-inch congress style safety shoe with Goodyear welt construction, cut from grain out upper leather, plain toe with elastic goring on each quarter, leather vamp lining and safety steel toe. The shoe shall have a non-marking, oil resistant rubber outsole and heel (see Figure 1).

3.2 Materials.

3.2.1 Upper leather (vamps and quarters). The uppers shall be cut from the best quality green-salted, chrome tanned sides either full grain or corrected grain, 4-1/2 to 6 ounces in thickness, an area of side not exceeding 28 square feet, and tannery lot not exceeding 25,000 square feet.
The grain side shall have a black finish and the flesh side shall be natural. The leather shall not be embossed or printed on. Leather components do not require a fungicidal treatment.

3.2.2 Insoles. The insole shall meet the following requirements in Table I. A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

TABLE 1 Insole Requirements

Characteristics	Physical Requirements	Test Method
Thickness	0.095 (+/- .005)	ASTM D-1777
Fungal resistance	Level 10	ASTM D-3273
Wet Tensile	70 kgf/cm^2 min.	SATRA PM 2
Flexing Index	3.7(5000 cycles)	SATRA PM 3
Abrasion Resistance Cellulose board Non-woven board	 $50mm^3$ max. 25,600 revs dry/6,400 revs wet	 SATRA PM 14 SATRA PM 31
Transverse tensile-Dry	8 kgf/cm^2 min.	SATRA PM 80
Dimensional Stability	0.7% max.	SATRA PM 98
Peel Strength	0.5 kgf/cm min.	SATRA PM 101

3.2.3 Stuck on rib strength. The physical requirement for the rib strength shall conform to Table II when tested as specified. A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

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Table II Rib strength requirments

Characteristics	Minimum 1/	Test method
Fabric Breaking Strength	200 lbs. (in fill direction)	ASTM-D-751
Shear Strength of Insole Ribs	85 lbf/inch	SATRA AM-5
Peel Resistance of Ribs Stuck to Insoles		
(Width dry)	3.1 lbf/rib	SATRA AM-5
(Width wet)	2.2 lbf/rib	SATRA AM-5

1/ No single determination (sample unit) shall fall below the minimum value specified and the average of all specimens (lot average) shall be not less than the average specified.

3.2.4 Vamp lining. The vamp lining leather shall be soft, pliable calf, kip, cattlehide, glove or pigskin leather of 2 to 3-1/2 ounces thickness, either chrome or vegetable tanned, full grain first split or partially corrected grain. The color shall be as produced by the tanning agent and a light application of dye may be used to level the color. When split pigskin is used, the pigskin shall be the first split from the grain side after the grain surface has been removed. The surface of the split nearest the grain surface shall be sealed with a clear lacquer and then smooth plated. The sealed plated surface shall be worn next to the foot.

3.2.5 Heel pads and inside counterpocket. The leather for the heel pads and inside counterpocket shall be soft, pliable calf, kip, cattlehide, glove or pigskin leather or sheepskin, kid, or calf, with a thickness of 2 to 3-1/2 ounces.

3.2.6 Welting, synthetic. The synthetic welting shall be flat top design conforming to dimensions of Figure 2, bright black in color, and formed with or without a storm welt. The welt thickness shall be 0.12 to 0.13 inch. The base compound shall be polyvinyl chloride. The welting shall conform to the requirements specified in Table III.

Table III. Synthetic welting requirements

Property	Requirements	Test method
Initial		
Hardness	85 – 95	ASTM-D-2240
Specific gravity	1.32 (max.)	see 5.2.2.1
Brittle point		
Before aging	-15°F or less	ASTM-D-746
After aging	+ 20°F maximum changes for standard brittle point	ASTM-D-746
Tensile strength, lb./sq.. in	2100 min.	ASTM-D-412
Elongation, percent	225 min.	ASTM-D-412

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Table III. Synthetic welting requirements (cont'd)

Property	Requirements	Test method
Water absorption, percent gain	3.00 max.	See 5.2.2.2
Solvent resistance (After immersion		
Hardness	55 min.	ASTM-D-2240
Tensile strength, lb./sq. in.	1800 min.	ASTM-D-412
Elongation, percent	255 min.	ASTM-D-412
Volume swell, percent	35 max.	ASTM-D-471

3.2.7 Counters. The counters shall be fabricated of either a full grain, vegetable tanned leather; leatherboard composed of fibrillated sole leather fibers bonded together with an elastomeric binder; or shoeboard composed of styrene butadine. In back part lasting operation, a synthetic material shall be required. The material shall be an extruded ionomer resin sheet material with a thickness of 0.060 ± 0.003 inches. The material shall contain a heat-activated adhesive on both sides with a coating thickness of 0.003 to 0.006 inch on each side. Any color resin shall be acceptable. The material shall conform to the requirements specified in Table IV.

Table IV. Counters requirement

Characteristic	Requirement	Test method 1/
Specific gravity	. 95 ± .015 g/cc	ASTM-D-792
Tensile strength	4000 ± 500 lb./sq. in.	ASTM-D-3574
Elongation	250 percent (min.)	ASTM-D-1708
Flex modulus	50,000 ± 5,00 lb./sq. in.	ASTM-D-790
Brittle temperature	-60° C (max.)	ASTM-D-746

3.2.8 Outsoles. The outsoles shall be 16 irons in thickness with a solid rubber composition, using Grade A rubber (see Figure 3). The rubber shall be black, non-marking, and conform to the requirements specified in Table V. As an alternate; the outsoles tread area may be of a straight -bar design or a commercial design.

TABLE V. Outsole requirements

Characteristic	Requirement	Test method
Abrasive index, minimum Before and after aging	150	ASTM-D-1630
Hardness Before and after aging		
Minimum	60	ASTM-D-2240
Maximum	80	ASTM-D-2240

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TABLE V. Outsole requirements (cont'd)

Characteristic	Requirement	Test method
Volume swell, percent, maximum	30	ASTM-D-471
Cut growth after aging Percent, maximum	250	ASTM-D-1052

3.2.9 Heels. The heels shall have a thickness of 7/32 (+ 2/32, - 1/32) inch, constructed with a plastic or composition core (see Figure 3) and conform to the requirements specified in Table VI. As an alternate, the heels tread area may be of a straight-bar design or commercial design.

TABLE VI. Heels requirements

Characteristic	Requirement	Test method
Stitch tear Before and after aging, Lb./inch, minimum	240 1/	ASTM-D-4786
After immersion Lb./inch. min	160	ASTM-D-4786
Tear strength, lb./inch, minimum	175	ASTM-D-624, Die C

1/ Aging shall be accomplished in accordance with ASTM-D-573. The soles and heels shall be aged 24 hours at 212° ± 1.8 ° F.

3.2.10 Fabrics.

3.2.10.1 Backseam tape. The backseam tape shall be 1/2 inch or 7/16 inch wide.

3.2.10.2 Pull strap. The front pull strap shall be 13/16 ± 1/16 inch wide black cotton tape with a minimum tensile strength of 70 pounds in the warp direction.

3.2.10.3 Goring. The goring shall be a woven elastic fabric, black in color, 4 to 4-1/2 inches wide for shoe sizes 8 and larger, and 3-1/2 to 4 inches wide for smaller sizes.

3.2.11 Boxtoe.

3.2.11.1 Understructure boxtoe. The understructure boxtoe shall be either an impregnated understructure, unicellular understructure , or laminated understructure. The impregnated understructure shall be made of two layers of single napped cotton fabric, finished thickness of

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0.034 to 0.038 inch, and minimum finished weight of 12-oz./sq. yd. The unicellular understructure shall be made of a closed cell (foam) material with a minimum thickness of 3/32 inch. The laminated understructure shall be made of ionomer resin laminated with a combination of flannel fabric on one side and nonwoven flannel or sheeting fabric on the other side with a combined thickness of 0.033 ± 0.005 inch.

3.2.11.2 Toe cushion. The material used to cushion the steel box toe and mask the breastline shall be a wool felt. As an alternate, the following materials may be used:

(a) Foamed polyvinyl material, 3/32 ± 1/32 inch in thickness weighing 4 to 6 pounds per cubic foot.
(b) Latex foam rubber 1/8 ± 1/64 inch thick.

3.2.11.3 Steel box toe. The steel box toe shall be fabricated from cold-rolled carbon steel. The steel box toes, after heat treatment, shall have a hardness of 43-50, carbon content of 0.50 to 0.82, thickness of 0.0625 ± 0.0025 inch. The finished boot shall meet or exceed the requirements of ANSI Z41, Class 75 for impact resistance and compression resistance. The tests shall be performed in accordance with the test methods cited under ANSI Z41.

3.2.12 Shank combination. The shank combination shall consist of a single or double rib steel shank attached by four prongs or two rivets to a shank board. As an alternate, a fiberglass shank without a shank board may be used.

3.2.12.1 Steel shank. The steel shank shall be 19 gage cold rolled carbon steel with a Rockwell C scale hardness of 47 to 54, width of 5/8 ± 1/64 inch, thickness of 0.080 to 0.125 inch, and zinc electroplated coating, dull or bright finish, or a zinc compound applied to the base metal.

3.2.12.2 Shank board. The shank board shall be water resistant with a finished thickness of 1/8 ± 1/32 inch. The shank board pieces shall be molded to follow bottom cavity of shoes, skived at both ends with a graduated scarf of 1-3/8 ± 1/16 inch and be graded up or down 1/4 ± 1/16 inch.

3.2.12.3 Fiberglass shank. The fiberglass shank shall consist of glass fibers impregnated with vinyl ester resin encased in a plastic sleeve or tube having a width of 5/8 ± 1/32 inch. The shank shall be cured and bonded to the insole and conform to the last shape.

3.2.13 Nails, heel attaching. Heel nails shall be brass, cut or wire type, and of sufficient length and gauge to produce a secure clinch on the insole.

3.2.14 Tacks and staples. Tacks and staples shall be brass or steel and of a sufficient length to thoroughly attach the parts.

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3.2.15 Nails, heel seat fastening. Nails used in heel seat fastening shall be a No. 12 through 39 head, brass nail or steel nail of sufficient length to secure parts.

3.2.16 Bottom filler. The bottom filler shall be either a thermoplastic type or cold process type. The thermoplastic type shall consist of a mixture of ground cork and a suitable thermoplastic binder. The cold process shall be spreadable without the use of heat and consist of a mixture of ground cork and a suitable binder.

3.2.17 Threads The upper stitching operations shall be performed using nylon thread. Thread size shall be Tex 80 and the color shall be black. Thread for Goodyear stitching and inseaming shall be polyester, Tex 800 and natural in color. Colorfastness requirements shall not apply for any of these threads.

3.2.18 Wax. The wax used during inseam sewing and Goodyear stitching shall be a permanently plasticized resin, white or golden in color.

3.2.19 Cleaner solution. The cleaning solution shall be a suitable one for removing grease and soil marks, have a maximum pH value of 8, and not decrease water-resistance of the upper leather.

3.2.20 Repairers and renovators. Repairs shall be liquid spray, crayon, and paste-type applied by hand. Renovators shall be sponge applied or sprayed. The color shall match the upper leather

3.2.21 Fillers. Fillers, when used, shall be applied by sponge or sprayed and shall be capable of providing a foundation for application of the top finish.

3.2.22 Top finish. Top finish shall be applied by either a sponge or spray method, capable of binding with previously applied coatings, and provide a flexible finish.

3.2.23 Adhesives. The adhesives used for bonding various parts of the shoes shall be either natural rubber latex, synthetic rubber latex, natural rubber solvent cement, synthetic resin cement, synthetic rubber solvent cement, or synthetic cement for sole stock fitting.

3.2.24 Removable insoles cushion insert. The cushion insoles shall be medium brown, open-celled polyurethane foam with an integral skin, laminated to a one iron impregnated cellulose substrate. The combined thickness shall be 0.130 $\pm$ 0.013 inch.

3.3 Patterns and lasts. Patterns and lasts shall be furnished by the contractor.

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3.4 Marking and labeling.

3.4.1 Marking. The contractor's name, number and date of contract and correct size and width shall be legibly stamped. Lettering shall be 1/8 to 1/4 inch high.

3.4.2 ANSI Labeling. The shoes shall be labeled in accordance with the Class 75 safety toe requirements specified in ANSI Z41.

3.4.3 Label/tag. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper tag shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each item by a fastener, clearly legible and readable by a scanner. The bar coding element shall be a 13 digit national stock number (NSN). The bar code type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and so that it causes no damage to the item.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Visual examination. Each shoe molder shall be examined for the defects listed below.

5.2.1 Defects. Not properly mated; any spot, stain, or foreign matter; color and finish not as specified; design, type, and size not as specified; any component not fabricated of the material specified; upper leather stretchy, damaged, or with breaks; lining leather that is wrinkled or excessively full; any cut, tear, hole, repair, abrasion; any component or assembly misplaced, improperly performed, or omitted; any open seam, loose or tight tension; wrong stitch or seam types; stitching omitted or not as specified; outsole not as specified; counter rolled or curled; edge making not properly trimmed or extended; thread ends not removed, or loose tension resulting in loose seams; any material defects, distorted parts or poor workmanship; insole wrong size; any protruding point, tack, staple or nail on insole; any component part omitted; label missing, incorrect, or illegible; measurement of item not as specified; bar code omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged item; bar code causes damage to the item; any items not packaged in accordance with the contract or purchase order.

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5.2.2 Tests

5.2.2.1 Specific gravity. The specific gravity for the synthetic welting shall be determined by a gravitometer using a specimen from 2 to 3 inches in length.

5.2.2.2 Water absorption. A specimen of synthetic welting shall be cut to a length of 4 ± 0.05 inches, weighed to the nearest 0.001 gram (W1), immersed in a beaker of distilled water. The top of the beaker shall be sealed during the test to present evaporation of the distilled water and maintained for seven days in an oven at a temperature of $70^{o} \pm 2^{o}C$. The beaker shall then be removed from the oven, cooled to a room temperature. The test specimen shall then be washed in methanol, dried for one minute and reweigh (W2). The percentage of water absorbed shall be calculated from the following formula:

$$\text{Water absorption} = \frac{(W2-W1}{(\ W1\ )} \times 100$$

5.3 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order.

6. PACKAGING

6.1 Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Source of Government documents. Copies of military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

7.2 Source of non-Government documents

ASTM-D-412	-	Rubber, Vulcanized and Thermoplastic Elastomers-Tension
ASTM-D-471	-	Rubber Property-Effects on Liquids
ASTM-D-573	-	Rubber-Deterioration in an Air Oven
ASTM-D-624	-	Rubber and Thermoplastics Elastomers, Tear Strength of Conventional Vulcanized
ASTM-D-746	-	Plastics and Elastomers by Impact, Brittleness Temperature Of

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ASTM-D-751	-	Fabrics, Coated
ASTM-D-790	-	Plastics and Electrical Insulating Materials, Flexural Properties of Unreinforced and Reinforced
ASTM-D-792	-	Density and Specific Gravity (Relative Density) of Plastics by Density
ASTM-D-1052	-	Rubber Deterioration-Cut Growth Using Ross Flexing Apparatus, Standard Specification for Measuring
ASTM-D-1630	-	Rubber Property- Abrasion Resistance (NBS Abrader)
ASTM-D-1708	-	Plastics by Use of Microtensile Specimens, Tensile Properties Of
ASTM-D-2240	-	Rubber Property – Durometer Hardness
ASTM-D-3273	-	Resistance to Growth of Mold on the Surface of Interior Coatings.
ASTM-D-3574	-	Materials, Flexible Cellular-Slab, Bonded, and Molded Urethane Foams
ASTM-D4786	-	Stitch Tear Strength, Single Hole

(Applications for copies should be addressed to American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428.)

ANSI Z41	-	American National Standard for Personal Protection-Protective Footwear

(Applications for copies should be addressed to the National Safety Council, P.O. Box 558, Itasca, IL 60143-0558.)

SATRA FOOTWEAR TECHNOLOGY CENTER

AM-5	-	Shear Strength of Insole Rib
PM-2	-	Tensile Strength of Fiberboard and Sheet Insoling Materials
PM-3	-	Flexing Endurance Test of Fiberboard
PM-14	-	Scuff Resistance of Fiberboard
PM-31	-	Martindale Abrasion
PM-80	-	Transverse or Z Direction Strength of Fiberboard
PM-98	-	Determination of Dimensional Change of Insole Boards Due To Change In Relative Humidity
PM-101	-	Peel Strength of Fiberboard

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(Application for copies of SATRA Test Methods should be addressed to:

Satra Technology Centre, Satra House
Rockingham Road, Kettering, Northamptonshire
NN 169 JH, England

7.3 Suggested manufacturers. In a limited market search for Shoe, Molders that meet the salient characteristics of this commercial item description, the following manufacturers are listed below. It must be noted that the list is not all-inclusive and that there may be other manufacturers in the market place with a suitable product.

Addison Shoe Company, Style Numbers ID- 6150 or ID-6296, Wynne, AR 72396
American West Shoe Company, Style Number 90045 Waverly, TN 37185
Cove Shoe Company, Stock Number 26703, Martinsburg, PA 16662
Wolverine/Bates Shoe Company Style Number 00125, Rockford, MI. 49351

MILITARY INTERESTS:

Custodians
Navy - NU
Army - GL

Review Activities
Army - MD
Navy - MC, AS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:
DLA - CT

Project 8430-0449

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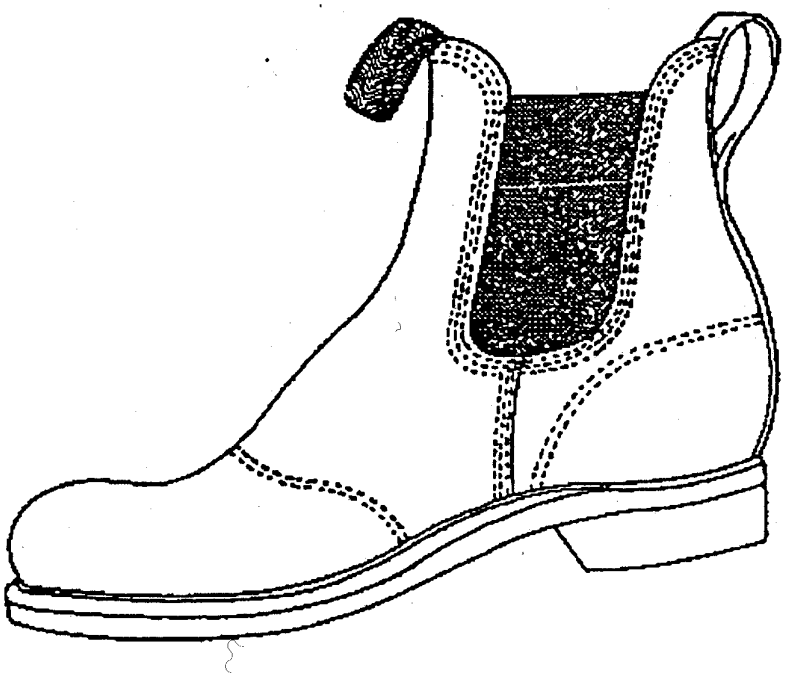
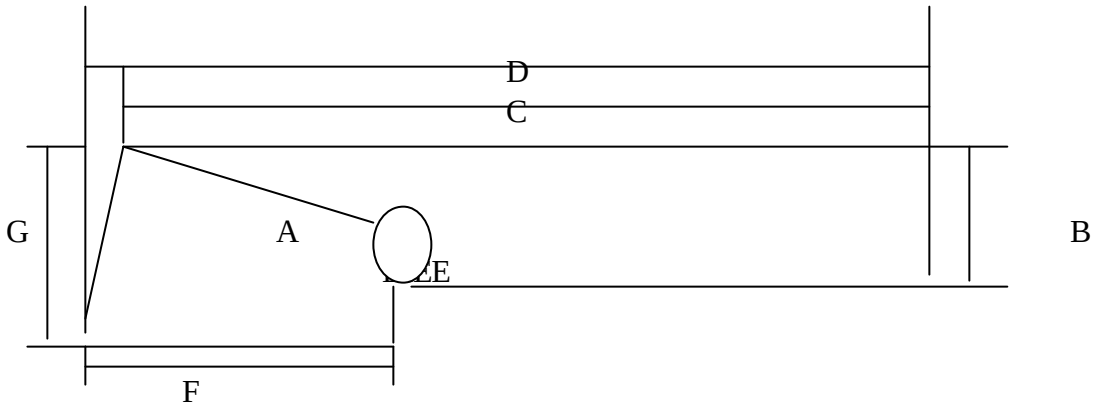


FIGURE 1. SHOE, MOLDERS

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A - .078 inches

B - .125 inches

C - .437 inches ± 1/128(.0078) inch

D - .456 inches

E - .060 inches

F - .109 inches

G - .188 inches

VINYL WELTING

FIGURE 2. SHOE, MOLDERS

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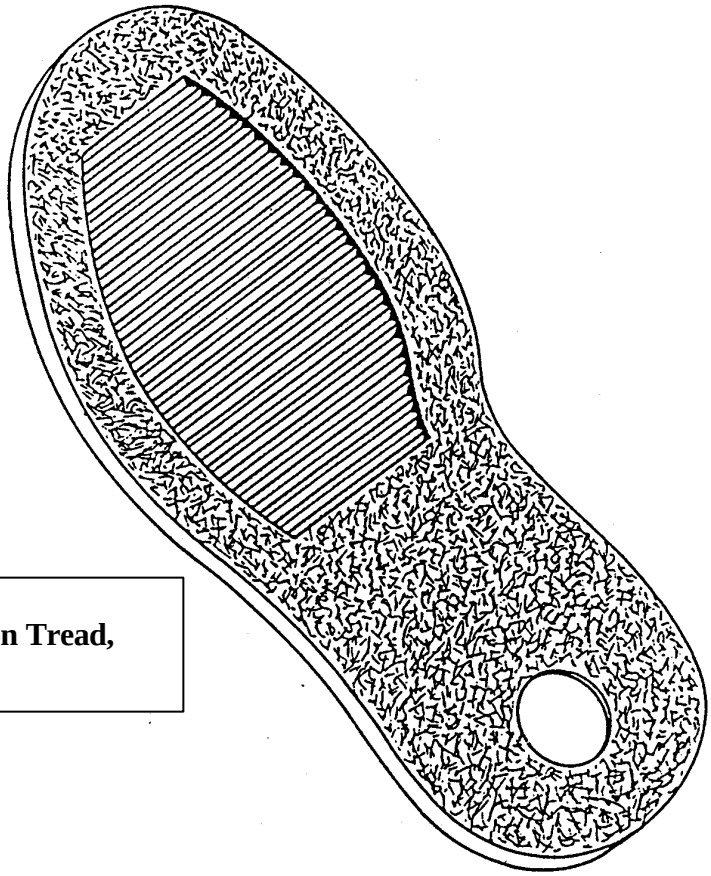


Figure 3
Soles, Rubber, Traction Tread, Shoes

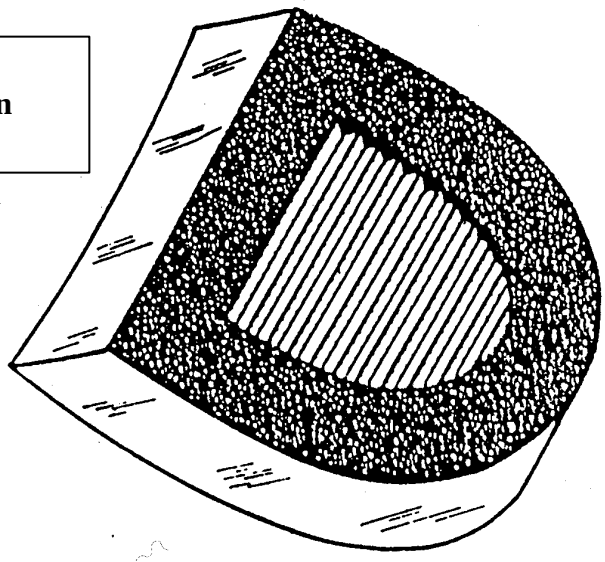


Figure 4
Heels, Rubber, Traction Tread, Shoes

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